ADS401 EVALUATION METHODS AND STATISTICS

Museum AI Guide: From Responses to Statistical Evidence

Presentation 3 • Data analysis and interpretation

Comparing post-visit experience with and without the museum's AI guide.

30 students · 15 per group · nine self-reported experience ratings.

GROUP H

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Study recap: a two-group experience comparison

RESEARCH QUESTION

How do post-visit experience ratings differ between students who used the museum's AI guide and those who did not?

Design & participants

Between-subjects comparison: AI guide n = 15; no AI guide n = 15.

Sample: 30 XJTLU students — 17 undergraduates and 13 postgraduates. All consented and had just completed a visit.

Conditions: the AI group used the museum guide; the comparison group visited without it. Both groups reportedly saw the same exhibition.

Procedure: consent → assigned/observed condition → museum visit → post-visit questionnaire.

Measures & design boundary

Q7–Q15: nine 1–5 ratings of engagement, interest, learning, understanding, reflection, meaning, context, emotion and theme resonance.

No verifiable primary endpoint was specified before analysis; all nine item comparisons are treated as exploratory.

Allocation was planned as alternating by arrival, but the recorded sequence is not strictly alternating. The study is therefore not labelled randomized or strictly alternating.

Dataset and preparation: 30 complete cases

One anonymised questionnaire file

16 variables: ID, consent, recent visit, AI-group indicator, role, prior visits, AI-tool frequency and nine experience items.

Coding: AI 1 = used guide, 0 = no guide; Q7–Q15 scored from 1 (strongly disagree) to 5 (strongly agree).

The clean CSV maps category labels to numeric codes; response values were not changed.

Quality checks

30 unique IDs (R01–R30); no duplicate records.

All 30 participants consented and confirmed a recent visit.

270 / 270 experience ratings are present and within the valid 1–5 range.

Consequently, no cases or item responses were excluded from the reported analysis.

Scope of evidence

The dataset supports item-level comparisons of self-reported experience. It contains no task-success, completion-time, behavioural-observation or SUS variables, and no validated total-score rule was supplied.

Analysis: nine exploratory two-group tests

1
Describe
Compute each item's mean, median and spread separately for the AI and no-AI groups.

2
Compare
Run a two-sided Mann–Whitney U test for each ordinal 1–5 item, with 15 participants per group.

3
Adjust
Apply Holm correction across the nine related tests; report both raw and adjusted p-values.

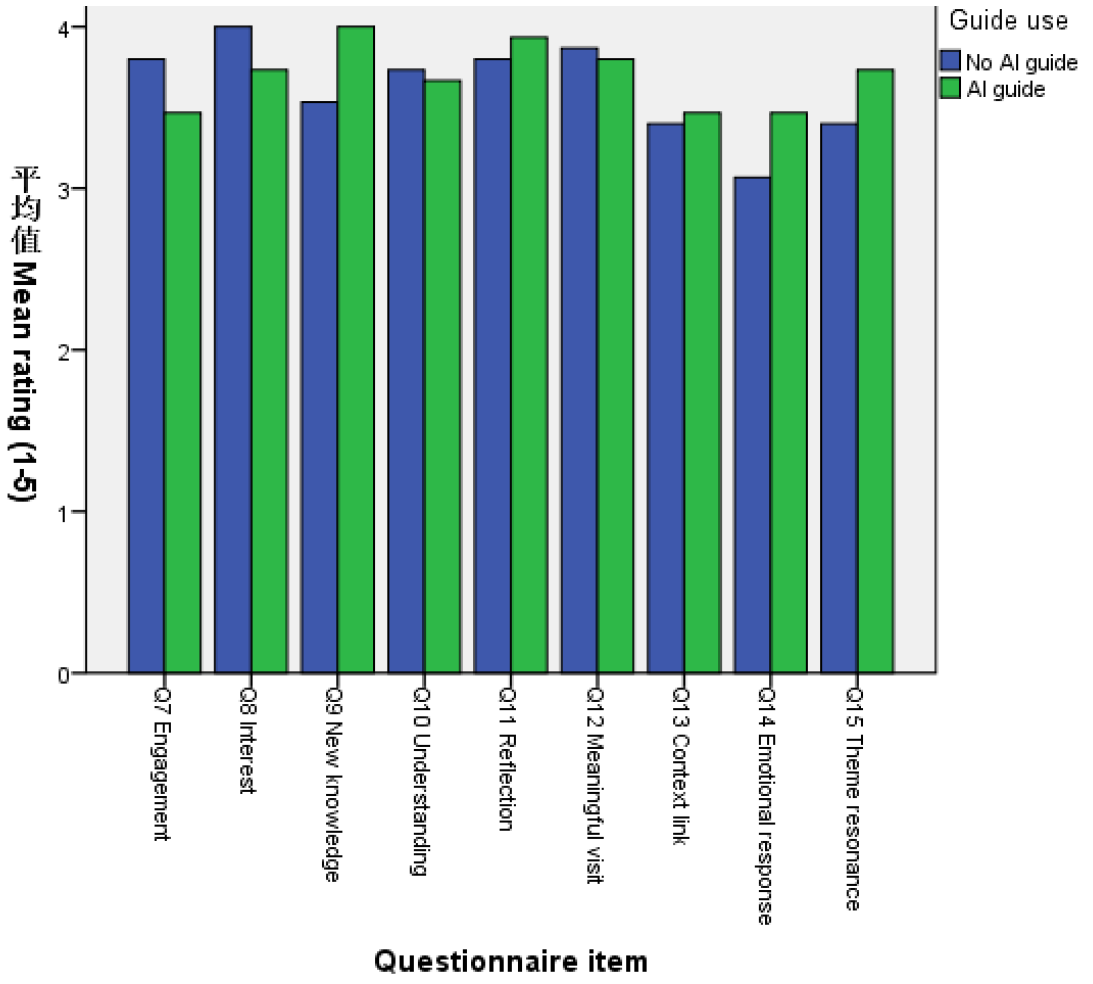
4
Quantify
Derive rank-biserial correlation from U to describe effect direction and magnitude.

Inference rules

H_0: no distributional difference between groups · two-sided $\alpha = .05$ · ties handled in the rank calculation.

Effect size: r_rb = 1 − 2U_no-AI / (15 × 15). Positive values favour the AI group; negative values favour the no-AI group. Means are descriptive; the tests use ranks.

Results: descriptive differences are small



What the bars show

AI mean higher on 5 items: Q9 +0.47, Q11 +0.13, Q13 +0.07, Q14 +0.40 and Q15 +0.33.

No-AI mean higher on 4 items: Q7 −0.33, Q8 −0.27, Q10 −0.07 and Q12 −0.07.

The largest positive difference is Q9 “learned new knowledge”, but its raw p = .230 and Holm-adjusted p = 1.000.

No item reached statistical significance before or after Holm correction.

Source: SPSS grouped mean chart, current _01 dataset. Bar height is descriptive; inference is based on ranks.

Results: all nine Holm-adjusted p-values = 1.000

Item	No AI mean	AI mean	Δ AI−No AI	U (no AI)	Raw p	Holm p	r_rb
Q7 Engagement	3.80	3.47	−0.33	138.0	.278	1.000	−0.227
Q8 Interest	4.00	3.73	−0.27	135.5	.328	1.000	−0.204
Q9 New knowledge	3.53	4.00	**+0.47**	84.5	**.230**	1.000	**+0.249**
Q10 Understanding	3.73	3.67	−0.07	118.5	.804	1.000	−0.053
Q11 Reflection	3.80	3.93	+0.13	105.0	.757	1.000	+0.067
Q12 Meaningful	3.87	3.80	−0.07	124.0	.622	1.000	−0.102
Q13 Context link	3.40	3.47	+0.07	108.5	.875	1.000	+0.036
Q14 Emotional response	3.07	3.47	+0.40	90.0	.318	1.000	+0.200
Q15 Theme resonance	3.40	3.73	+0.33	94.0	.427	1.000	+0.164

n = 15 per group; two-sided Mann–Whitney tests. Δ is the difference of group means. Positive r_rb favours AI; all effect magnitudes are below .30. Q9 has the smallest raw p-value, but remains non-significant.

Conclusion: no evidence of a difference in this sample

Answer to the research question

The current data do **not provide sufficient evidence** that using the museum's AI guide changes any of the nine self-reported experience ratings.

This is **not proof of equivalence or no effect**. With only 15 participants per group, the study has limited ability to detect modest differences.

Q9 is hypothesis-generating only: AI − no AI = +0.47, r_rb = +0.249, raw p = .230, Holm p = 1.000.

Next study: verify allocation, pre-specify one primary outcome, recruit a larger sample and retain a preregistered analysis plan.

Limitations & reflection

The allocation record is inconsistent with strict alternation; causal attribution to AI is therefore unjustified.

A small convenience sample of students at one museum limits generalisability and statistical power.

Outcomes are self-reports only; nine comparisons and no verified primary endpoint make the analysis exploratory.

Methodological lesson: transparent group records, version control and correction for multiple testing are as important as the test itself.

Q&A

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